

Use this document to detail specifications/requirements for an item and/or ancillary services. ⓘ

TITLE: 1550 nm Pulsed Fiber-coupled Laser ⓘ

Requesting Lab/Division/Group: Physical Measurements Lab, Applied Physics Division,
Sources and Detectors Group

Requisition Number: NB686010-26-01534

General Statement of Need ⓘ

The Government requires a fiber-coupled pulsed laser source operating at 1550 nm for laboratory use. The required system must provide a pulse width somewhere between 500 fs to 50 ps, support repetition-rate control up to 80 MHz and down to 1 Hz, provide at least 1 mW average output power, use single-mode fiber with an FC/APC connector, and support USB and/or GPIB control without relying on Ethernet as the sole control interface.

Items and/or Ancillary Services ⓘ

Item 1: 1550 nm Pulsed Fiber-coupled Laser System

Quantity: 1

Specifications:

1. The system must provide optical pulses at a nominal operating wavelength of 1550 nm.
2. The system must provide pulse duration within the range of 500 fs to 50 ps. The pulse duration does not need to be tunable.
3. The system must provide an external trigger input.
4. The system must provide an output electrical trigger timed with the output optical pulses.
5. The background optical power must be 30 dB below the optical pulse power.
6. The system must provide user-selectable repetition rate up to at least 80 MHz and down to 1 Hz. If operation at the low end of the range is achieved through pulse picking, external triggering, or an included accessory, that capability shall be included in the quoted configuration.
7. The system must provide at least 1 mW fiber-coupled average optical output power at 1550 nm under the quoted operating condition(s) and no more than 300 mW.
8. The output shall be fiber coupled.
9. The output fiber shall be single-mode fiber.
10. The optical output connector shall be FC/APC.
11. The system shall support remote control and/or instrument communication through USB and/or GPIB.
12. The system shall not require Ethernet as the only remote-control interface.
13. The vendor shall identify all accessories, pulse-picking modules, trigger options, software, and control hardware required to satisfy the stated pulse-width and repetition-rate requirements.
14. The vendor shall provide documentation sufficient to verify compliance with the quoted pulse duration, repetition rate, optical output power, fiber type, connector type, and control interface.
15. The delivered system shall be suitable for laboratory use and shipped ready for operation

with the quoted fiber-coupled output configuration.

Separately Priced Options and Services:

Item 2: [List minimum specifications.

Quantity: [Enter quantity here.]

Specifications: [Enter quantity here.]

Click or tap here to enter text.

Service Requirements / Item Specifications

Section ⓘ	Requirements/Specifications ⓘ
Planning Considerations ☒ N/A	☐ List any visits required to allow potential vendors to see site specific constraints. [Explain requirement here] ☐ US citizen required on-site (foreign nationals require preregistration 30 days prior) ☐ Vendor rep on-site more than three days ☐ Other considerations: [List other considerations to ensure an accurate quote]
Shipping & Delivery ☐ N/A	Standard shipping Shipping criteria: ☐ Partial delivery acceptable ☐ Direct delivery to building required [Explain need for requirement here] Building has: ☐ Loading dock ☐ Containerization preference: [Choose container] Other requirements: [List any other requirements here] Delivery date criteria: ☐ Delivery of goods, installation, and training shall be completed no later than: 08 Weeks ARO ☐ Deliver no earlier than: [Insert date here] ☐ Other: [List other shipping, delivery, or specil requirements]
Electronic Media ☒ N/A	[Enter electronic media requirements by item here.] ☐ Software ☐ Electronic manuals ☐ License Key ☐ Other: [List any other Electronic Media requirements here] [Enter email address for software delivery here]
Installation ☒ N/A	[Enter installation requirements by item here.] ☐ Vendor technician on site ☐ Rigging ☐ Uncrating / Unpackaging ☐ Removal of packaging material ☐ Equipment set up ☐ Start-up services

Requirements / Specification Document

	☐ Turn key installation ☐ Other [Enter details here.] Note: When installation is required, TPOC or COR shall be the coordination point.
Facility / Utility Considerations ☒ N/A	No utility or facility requirements. ☐ OFPM Work Order Number(s) [List any Work Order Number(s) here] ☐ Power available Voltage: [Enter voltage here] (V) Max Current: [Enter max current draw here] (A) Phase: [Choose phase] Frequency: [Typically 60Hz] (Hz) NEMA Plug Type: [Enter plug type here] (i.e.: NEMA L5-20P) Environment: [Choose environment] Other power consideration: [List any other power considerations here] ☐ Utility connections: [List relevant available utilities, pressure(s), and flowrate(s)] ☐ Other considerations: [List other considerations to ensure an accurate quote]
Warranty ⓘ ☒ N/A	Base Manufacturer Warranty is acceptable [If you require an Extended Manufacturer Warranty, describe the warranty requirements per item here.] What is included? ☐ Labor ☐ Parts ☐ Travel ☐ Cost and liability for returns ☐ Other: [Enter other requirements here] ☐ Warranty length: [Enter warranty length here] Period Length:[Ex. month, 6 month, year etc.]
Training ☒ N/A	No training requirements ☐ On-site ☐ Off-site location: [Enter location(s) here] ☐ Number of trainees: [Enter number of trainees here] Scope of training: ☐ Operations ☐ Maintenance ☐ Troubleshooting ☐ Safety ☐ Other: [Enter other scope requirements here] ☐ Training materials to be provided: [Describe requirement here] ☐ Other: [Enter other training requirements here] Note: When training is required, TPOC or COR shall be the coordination point.
Inspection and Acceptance ☐ N/A	[Enter inspection requirements per item here] ☐ Planned government acceptance testing (7 business days) ☒ Gov't acceptance expected to take longer than 7 business days: 14 days

	☐ Inspection and acceptance plan: [Explain here] ☐ Other: [Enter other requirements here]
Maintenance ⓘ ☒ N/A	none Period Length: [Ex. month, 6 month, year etc.] What is included? ☐ Labor ☐ Spare parts ☐ Consumables ☐ Travel ☐ Cost and liability for returns ☐ Software updates ☐ Manuals ☐ Service plan length: [Enter service plan length here] ☐ Other: [Enter other requirements here] ☐ Requirements: [Enter requirements here] (i.e.: onsite within 72 hours, etc.) ☐ Option Period 1 ☐ Option Period 2 ☐ Option Period 3 ☐ Option Period 4
Government Furnished Material / Equipment ☒ N/A	none ☐ Samples for testing ☐ Government equipment which will leave gov't site (TPOC must coordinate with division property custodian) Serial Number: [Enter Serial Number here] Description: [Enter description here] NIST property number: [Enter NIST property number here] ☐ Other: [Enter other details here]
Travel ☒ N/A	none ☐ Explain: [Explain here] ☐ Other: [Enter other requirements here] <i>Vendor: To the maximum extent possible, include as part of appropriate sections above (maintenance, repair, etc.). Otherwise, it will be a separate line item, reimbursable per the Federal Travel Regulation.</i>

Provide suggestions to improve this form by clicking [here](#).